

Learning-Map Tour — ELF: Embedded Language Flows

Tour artefact (v1.9 per-paper-tour rollout)

Hu et al. 2026, [arXiv:2605.10938](#)

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1. Reader’s Contract

Audience. Math-grad to ML researcher. The reader is comfortable with measure-theoretic probability, ODEs/SDEs at the level of a first graduate course, and basic Riemannian geometry; familiarity with Transformers and language modeling is assumed but not deep expertise in diffusion-language work.

Background. Before reading the paper itself, walk the [math-foundations](#) tour at the *Graduate* entry point. The reader who has not internalised continuity equations and Liouville flow on densities should warm up there first. Foundation links are given concept-by-concept in Section 2.

Expected reading time. 3 to 4 hours, broken as:

- 30 min — foundations refresher (Section 2 of this tour)
- 90 min — first read of `02-math-deep-dive.md` end-to-end
- 30 min — `05-improvements.tex` (the proposed extensions and their proofs)
- 60 min — running `cfg-schedule.py`, sketching an experiment, writing notes in `03-opinions.md`

Elevator (one sentence). ELF runs continuous-time flow matching in a *frozen pretrained embedding space* and reuses *the same network’s final timestep* as the decoder via the *weight-tied unembedding matrix*.

Why it matters. The headline win is structural, not just empirical: discrete-token diffusion models (MDLM, SEDD) and continuous-noise diffusion language models had been treated as separate research programs. ELF shows that the second program, done in the right embedding space and decoded by the same network, dominates the first while shedding the separate-decoder overhead.

2. Foundations Walk

Topologically ordered. The pacing tag (SKIM, READ, DRILL) signals depth.

#	Concept	Pacing	Why ELF needs this
06	Linear Maps and Matrices	skim	Token embedding W_E and its transpose are the load-bearing linear maps; weight-tying says they are the same matrix.

#	Concept	Pacing	Why ELF needs this
14	Gradient and Jacobian	read	The velocity field $v_\theta(z_t, t)$ is regressed against a Jacobian-defined target; gradient flow on the FM loss is the optimisation story.
24	Probability Density Functions	read	$z_0 \sim \mathcal{N}(0, I)$ and $z_1 \sim p_{\text{data}}$ are densities on $\mathbb{R}^d$; ELF is a transport between them.
25	Expectation	skim	The flow-matching loss is an expectation over (t, z_0, z_1) .
28	Change of Variables (Probability)	drill	The continuity equation $\partial_t p_t + \nabla \cdot (p_t v_t) = 0$ is the infinitesimal change-of-variables that makes flow matching well-defined.
29	Ordinary Differential Equations	read	Inference is the deterministic ODE $dz_t/dt = v_\theta(z_t, t)$, integrated from $t = 0$ to $t = 1$ with Euler or Heun.
32	Brownian Motion / Wiener Process	skim	Background; continuous-time noise diffusion (ELF’s lineage) is built on Brownian motion.
33	Stochastic Differential Equations	skim	DDPM/SEDD are SDEs; flow matching is the deterministic ODE limit. Side-by-side clarifies what ELF gives up and gains.
37	Cross-Entropy	read	Final-step loss is cross-entropy between $W_E^\top z_1$ (logits) and the true token, tying the continuous flow to a discrete objective.
38	KL Divergence	skim	CFG re-derives as a KL-regularised posterior interpolation; useful framing for $\omega > 1$.
40	Information Geometry (Fisher Metric)	drill	The proposed curved-path improvement lives on a Riemannian manifold whose metric is the Fisher information at each embedding.
41	Optimal Transport (Wasserstein)	drill	Linear interpolation is the Euclidean W_2 geodesic; the improvement asks whether the true OT geodesic on the embedding manifold differs.

3. Paper Concepts Walk

ELF-specific moving parts, lifted directly from 02-math-deep-dive.md.

Concept	What it does in the paper	Reference
Linear interpolation path	Defines the trajectory $z_t = (1 - t)z_0 + tz_1$ along which the velocity target is computed; the constant $z_1 - z_0$ is the regression target.	Eq. (1), Setup
Velocity-field prediction	Network v_θ trained against $\mathcal{L}_{\text{FM}} = \mathbb{E} \ v_\theta(z_t, t) - (z_1 - z_0)\ ^2$.	Eq. (2), Central Object

Concept	What it does in the paper	Reference
Weight-tied unembedding	At $t = 1$, logits = $W_E^\top z_1$ with the same W_E used at encode-time. No separate decoder.	Final-Step Discretization
Classifier-free guidance	$v_{\text{cfg}}(z_t c) = \omega v_\theta(z_t c) + (1 - \omega)v_\theta(z_t \emptyset)$, imported from image diffusion.	Eq. (4)
Self-conditioning	The conditioning c is the model’s own intermediate prediction; CFG without an external conditioner.	Self-Conditioning
Training-time CFG	Network is trained to output v_{cfg} directly in one forward pass, halving inference cost.	Eq. (4) discussion
Pretrained contextual embeddings	W_E initialised from a frozen pretrained encoder; ablations show non-contextual or random embeddings collapse the lift.	Embedding Choice
Latent-diffusion ancestry	ELF is Latent Diffusion specialised: encoder = embedding matrix, decoder = its transpose.	Connections

4. Improvements Walk

Each entry in `05-improvements.tex` is graded **PROOF** (derivation in a `.tex`) or **MEASUREMENT** (numbers from a Python script). Deferred items lack compute and are flagged.

#	Type	Title	Artefact	Status
Math 1	PROOF	When is the linear path optimal?	<code>proofs/curved-path-optimality.tex</code>	live
Math 2	MEASUREMENT	Decomposing embedding-quality vs flow-formulation	counterfactual training deferred run	deferred
Code 1	MEASUREMENT	Scheduled CFG: $\omega(t)$ rather than constant ω	<code>improvements/cfg-schedule.py</code> (extended with <code>measure()</code>)	live
Code 2	MEASUREMENT	Multimodal embedding sharing prototype	<code>improvements/multimodal-embed-schedule.py</code>	deferred
Exp 1	MEASUREMENT	VR-baseline-controlled comparison at matched compute	105M decoder-only AR deferred run	deferred
Exp 2	MEASUREMENT	Multimodal embedding sharing experiment	<code>improvements/multimodal-embed-schedule.py</code> (live only)	live
Theory 1	PROOF	ELF as Latent Diffusion with weight-tied decoder	<code>proofs/elf-as-LDM.tex</code>	live
Theory 2	PROOF	ELF as discrete information bottleneck	<code>proofs/elf-info-bottleneck.tex</code>	deferred

Pacing recommendation. Read Math 1 and Theory 1 in full; they are tractable and clean. Run `cfg-schedule.py` interactively — the toy result is the only piece of original-data evidence in this study and is the most interview-portable artefact. Glance at Theory 2 and the deferred experiments; they are framed as “what would we measure if we had the GPUs.”

5. What To Do Next

1. **Most promising proposal — scheduled CFG.** The U-shaped $\omega(t) = \omega_{\max} (2t(1 - t))^{-\beta}$ schedule (Code Improvement 1) is concrete, prototyped on a 2D toy, and requires *no retraining* to test on a real ELF checkpoint — only inference-time modification. Reach out to the ELF authors with the schedule and the toy result.
2. **Single most valuable experiment — AR baseline at matched compute.** Train a 105M-parameter decoder-only Transformer on the same OpenWebText corpus for the same number of tokens as ELF’s headline checkpoint. The single missing row of Table 1.
3. **Follow-on paper direction — “How much of ELF is the manifold, how much is the flow?”** Combine Math Improvement 2 (counterfactual: discrete diffusion on the same pretrained embedding space) with the scheduled-CFG result. The paper writes itself: *ELF’s lift decomposes into $X\%$ from the embedding manifold and $Y\%$ from the flow; here is the right CFG schedule for both regimes.*